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ELGILOY SPECIALTY METALS

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[Elgiloy Specialty Metals](#) > Wire – Elgiloy Alloy

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[Back to Wire Alloys](#)

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Elgiloy® alloy wire

UNS R30003

W.Nr 2.4711

Applicable Specifications

Wire & Bar: AMS 5833, AMS 5834, ASTM F1058, NACE MR0175 (ISO 15156-3), ISO 5832-7, UNS R30003

Elgiloy® alloy Description

Elgiloy® alloy is a Cobalt-Chromium-Nickel-Molybdenum alloy having a unique combination of very high strength, excellent corrosion resistance, and high fatigue strength. Elgiloy® alloy wire is highly resistant to sulfide stress corrosion cracking and as a result, is the alloy of choice in the most demanding sour well environments. In

seawater Elgiloy[®] is virtually immune to crevice and stress corrosion and pitting even at the highest strength levels and can operate at cryogenic temperatures up to 850°F (454°C).

Applications

Springs, torsion bars, seals, medical devices & implants and orthodontic wire.

Industries Supplied

Oil & Gas Extraction, Medical & Dental, Aerospace, Defense and Space Exploration.

Nominal Composition

Co: 39.0 – 41.0
Cr: 19.0 – 21.0
Ni: 14.0 – 16.0
Mo: 6.0 – 8.0
Mn: 1.5 – 2.5
Fe: balance
Si: 1.2 max
C: 0.15 max
Be: 0.1 max
P: 0.015 max
S : 0.015 max

Physical Properties

Density: 0.300 lb/in³, (8.30 g/cm³)

Modulus of Elasticity (E):

At 70°F (20°C): Up to 32.0 x 10³ ksi (190 GPa)

Modulus of Rigidity (G):

At 70°F (20°C): 11.2 x 10³ ksi (77.2 GPa)

Coefficient of Expansion:

8.4 μin/in.-°F (70°F to 600°F)

15.2 μm/m-°C (20°C to 300°C)

Electrical Resistivity: 39.2 μΩ.in, (99.6 μΩ.cm)

Thermal Conductivity: 86.5 Btu-in/ft²hr-°F, (12.5 W/m-K)

Typical Mechanical Properties – Spring Applications

- **Annealed**
 - **Heat Treatment:** 2048°F (1120°C)
 - **Tensile Strength:** 120 – 150 ksi; (830-1035 MPa)
 - **Suggested Operating Conditions:** -300°F to 850°F (-184°C to 454°C)
- **Spring Temper**
 - **Tensile Strength:** 220 – 290 ksi; (1515 – 2000 MPa)
 - **Suggested Operating Conditions:** -300°F to 850°F (-184°C to 454°C)
- **Spring Temper + Aged**
 - **Heat Treatment:** After spring coiling. Age: 980°F (527°C) for 5 hours.
 - **Tensile Strength:** 260– 330 ksi; (1790 – 2275 MPa)
 - **Suggested operating conditions:** -300°F to 850°F (-184°C to 454°C)

Elgiloy[®] is a registered trademark of Elgiloy Specialty Metals.

For further information Contact:

Contact Us

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ESM is accredited / approved by



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WIRE DRAWING

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